

IMAT MOCK EXAM 6 — "At-Level Calibration"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 45/90.0 · Mock exam

Section A - Logical Reasoning & Reading (Q1-9)

1. "Bacteria store fragments of viral DNA in ordered arrays (CRISPR loci) in their own genome. These fragments are transcribed into guide molecules used by Cas proteins, which cut any invading DNA whose sequence matches a guide. A virus escapes this defence only if it mutates the exact targeted sequence. Because the array is inherited, daughter cells of a survivor remain protected."

Which of the following is not supported by the passage?

- A) A bacterium that survives infection can pass protection on to its offspring.
- B) The defence is guided by RNA copies of stored viral sequences.
- C) Mutation of the targeted sequence can let a virus evade the defence.
- D) During infection, bacteria deliberately rewrite their own DNA to match the invading virus.
- E) The system behaves as a form of inherited immunity.

2. Five friends P, Q, R, S and T sit in a row of five seats, all facing the same direction, with seat 1 at the left end and seat 5 at the right end. Q sits immediately to the right of P. R does not sit adjacent to P. S sits at an end seat. T sits somewhere to the left of S.

Which of the following must be true?

- A) R sits between P and T.
- B) P sits at the far left.
- C) Q is not adjacent to R.
- D) T is not at an end seat.
- E) S sits at seat 5 (the far right).

3. (General knowledge) The 2026 Winter Olympic Games, held in February 2026, are hosted by:

- A) Milano and Cortina d'Ampezzo, Italy
- B) Venice, Italy
- C) Turin, Italy
- D) Palermo, Italy
- E) Rome, Italy

4. No reptile is warm-blooded. Some reptiles are kept as pets.

Which conclusion necessarily follows?

- A) No pet is a reptile.
- B) Some pets are warm-blooded.
- C) Some pets are not warm-blooded.
- D) Some warm-blooded animals are kept as pets.
- E) All pets are cold-blooded.

5. A clock shows 4:40. What is the angle between the hour hand and the minute hand?

- A) 40° B) 60° C) 100° D) 120° E) 260°

6. An enzyme was assayed at three pH values. Without inhibitor, the measured rates at pH 5, 7 and 9 were 20, 100 and 20 respectively. With a fixed concentration of inhibitor I present, the rates were 18, 50 and 18 at the same three pH values.

Which conclusion is best supported by the data?

- A) The enzyme is inactive at pH 9.
- B) The enzyme is most active at pH 7, and the inhibitor's relative effect is also greatest at pH 7.
- C) The enzyme's optimum lies between pH 7 and pH 9.
- D) The inhibitor reduces the rate by the same fraction at every pH.
- E) The inhibitor destroys the enzyme at acidic pH.

7. A large national audit reports that patients operated on in city hospitals have fewer post-surgical complications than patients operated on in small rural hospitals. The authors conclude that the quality of surgical care is worse in rural hospitals.

Which of the following, if true, most weakens the authors' conclusion?

- A) Rural hospitals carry out a higher proportion of emergency operations, which have higher complication rates in any setting.
- B) Rural surgeons work longer shifts than city surgeons.
- C) Some city hospitals are also teaching hospitals.
- D) City hospitals have more beds than rural hospitals.
- E) City hospitals have newer pieces of equipment on average.

8. A sequence begins 2, 4, 7, 14, 17, 34, ... following a repeating two-step rule. What is the next term?

- A) 35 B) 37 C) 51 D) 68 E) 71

9. A jacket costing 120 is discounted by 20%, and the new price is then discounted by a further 25%. What is the total percentage reduction from the original price?

- A) 35% B) 40% C) 42.5% D) 45% E) 46%

Section B - Biology (Q10-32)

10. In a dihybrid cross $AaBb \times AaBb$ with two unlinked genes (complete dominance at both), what is the probability that an offspring has genotype $AaBb$?

- A) $1/16$ B) $1/4$ C) $1/2$ D) $3/16$ E) $9/16$

11. Which cross is expected to produce a phenotypic ratio of 9:3:3:1 (two traits, complete dominance)?

- A) $Aa \times Aa$ B) $AABB \times aabb$ C) $AABb \times AaBB$ D) $AaBb \times AaBb$ E) $AaBb \times aabb$

12. A mother is a carrier of red-green colour blindness ($X^H X^h$) and the father has normal vision ($X^H Y$). What fraction of their sons is expected to be colour-blind?

- A) $3/4$ B) $1/4$ C) all of them D) 0 E) $1/2$

13. DNA replication is described as "semiconservative" because each daughter double helix:

- A) conserves half of the parent's genes
- B) conserves only the sugar-phosphate backbone of the parent
- C) is made entirely of new nucleotides
- D) contains randomly mixed old and new nucleotides within each strand
- E) contains one intact parental strand and one newly synthesised strand

14. The enzyme that synthesises an RNA strand from a DNA template during transcription is:

- A) helicase
- B) DNA polymerase
- C) reverse transcriptase
- D) RNA polymerase
- E) primase

15. Which nucleic-acid molecule physically carries the anticodon?

- A) small nuclear RNA
- B) mRNA
- C) tRNA
- D) rRNA
- E) the DNA template strand

- 16.** The α -helix of a protein is stabilised primarily by:
- A)** hydrogen bonds between backbone groups
 - B)** hydrophobic cores
 - C)** ionic bonds between consecutive amino acids
 - D)** peptide bonds between side chains
 - E)** disulfide bridges
- 17.** A point mutation changes an mRNA codon from AUG (methionine) to UGA. UGA is a stop codon. The most likely consequence is:
- A)** no change in the protein
 - B)** a frameshift downstream
 - C)** substitution of one amino acid by another
 - D)** premature termination, giving a truncated protein
 - E)** an elongated protein
- 18.** In aerobic respiration, CO_2 is released:
- A)** on the inner mitochondrial membrane, during electron transport
 - B)** in the cytosol, during glycolysis
 - C)** in the mitochondrial matrix, during pyruvate oxidation and the Krebs cycle
 - D)** in the intermembrane space, during proton pumping
 - E)** by ATP synthase
- 19.** Cyanide blocks complex IV of the electron transport chain. Why does ATP synthesis by ATP synthase then stop?
- A)** Electron transfer stops, protons are no longer pumped, and the gradient that powers ATP synthase dissipates.
 - B)** ATP synthase can run without a gradient for only a few minutes.
 - C)** ATP synthase is directly inhibited by cyanide.
 - D)** Glycolysis stops, so there is no pyruvate.
 - E)** The Krebs cycle accelerates and floods the matrix with protons.
- 20.** The main purpose of the fermentation step (lactic acid or alcoholic) in anaerobic conditions is to:
- A)** eliminate CO_2 from the cell
 - B)** synthesise NADH for the electron transport chain
 - C)** produce additional ATP beyond glycolysis
 - D)** remove glucose from the cell
 - E)** regenerate NAD^+ so glycolysis can continue
- 21.** Two inhibitors were characterised for an enzyme. With inhibitor I, the apparent K_m is $4\times$ higher and V_{max} is unchanged. With inhibitor II, K_m is unchanged and V_{max} falls to a quarter. Inhibitor I is best classified as:
- A)** non-competitive
 - B)** uncompetitive
 - C)** an allosteric activator
 - D)** competitive
 - E)** an irreversible poison
- 22.** A cell's ATP is rapidly depleted by a metabolic poison. Which transport process will fail first?
- A)** water movement through aquaporins
 - B)** diffusion of oxygen through the membrane
 - C)** osmosis
 - D)** facilitated diffusion of glucose down its gradient
 - E)** Na^+/K^+ pump activity
- 23.** The neurotransmitter released at the neuromuscular junction of vertebrates is:
- A)** acetylcholine **B)** serotonin **C)** dopamine **D)** GABA **E)** adrenaline

- 24.** The resting membrane potential of a typical vertebrate neurone is closest to:
A) –70 mV **B)** –55 mV **C)** –30 mV **D)** 0 mV **E)** +30 mV
- 25.** Vitamin K is required in the liver for the synthesis of:
A) insulin
B) albumin only
C) prothrombin and other clotting factors
D) bile salts
E) haemoglobin
- 26.** After 12 hours without drinking, blood plasma osmolality rises. Which endocrine response helps conserve water?
A) aldosterone secretion increases
B) ADH (vasopressin) secretion increases
C) insulin secretion increases
D) atrial natriuretic peptide secretion increases
E) cortisol secretion stops
- 27.** In skeletal muscle contraction, which event directly exposes the myosin-binding sites on actin?
A) calcium binding to tropomyosin
B) sodium entering the sarcoplasmic reticulum
C) calcium binding to troponin
D) ATP binding to myosin heads
E) acetylcholine binding to the sarcolemma
- 28.** Antibodies circulating in plasma are produced by:
A) platelets
B) cytotoxic T cells
C) T helper cells
D) plasma cells differentiated from B lymphocytes
E) macrophages
- 29.** A human cell has just completed DNA replication (S phase). Which chromosome/chromatid count is correct?
A) 23 chromosomes, 23 chromatids
B) 46 chromosomes, 46 chromatids
C) 46 chromosomes, 92 chromatids
D) 92 chromosomes, 92 chromatids
E) 92 chromosomes, 46 chromatids
- 30.** The walls of the pulmonary alveoli are lined by:
A) ciliated columnar epithelium
B) simple squamous epithelium
C) transitional epithelium
D) stratified squamous epithelium
E) simple cuboidal epithelium
- 31.** The light-independent reactions of photosynthesis (Calvin cycle) take place in the:
A) thylakoid lumen
B) cytosol
C) thylakoid membranes
D) mitochondrial matrix
E) stroma of the chloroplast

32. Yeast cells are switched from aerobic to strictly anaerobic conditions. Which process can still produce ATP?

- A)** fatty-acid β -oxidation feeding the electron transport chain
- B)** glycolysis
- C)** proton flow through ATP synthase
- D)** the Krebs cycle
- E)** oxidative phosphorylation

Section C - Chemistry (Q33-47)

33. What is the oxidation number of chromium in the dichromate ion, $\text{Cr}_2\text{O}_7^{2-}$?

- A)** +3 **B)** +5 **C)** +6 **D)** +7 **E)** +12

34. In the reaction $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$, the reducing agent is:

- A)** Zn **B)** H^+ **C)** Cl^- **D)** H_2 **E)** Zn^{2+}

35. Which molecule has a net dipole moment (is polar overall)?

- A)** CO_2 **B)** BF_3 **C)** CCl_4 **D)** CH_3Cl **E)** CH_4

36. 0.001 M hydrochloric acid (pH 3) is diluted tenfold with water. The pH of the new solution is:

- A)** 2 **B)** 3 (unchanged) **C)** 4 **D)** 5 **E)** 7

37. What is the pH of a 0.01 M solution of NaOH at 25 °C? ($K_w = 1.0 \times 10^{-14}$)

- A)** 2 **B)** 10 **C)** 11 **D)** 12 **E)** 13

38. 3 mol of H_2 are mixed with 2 mol of O_2 and sparked: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. What is the maximum amount of water formed?

- A)** 1.5 mol **B)** 2 mol **C)** 3 mol **D)** 4 mol **E)** 6 mol

39. 20 mL of a 1.5 M HCl solution is diluted with water to a final volume of 150 mL. The concentration of the diluted solution is:

- A)** 0.10 M **B)** 0.15 M **C)** 0.20 M **D)** 0.25 M **E)** 0.30 M

40. How many molecules are present in 0.50 mol of CO_2 ? ($N_a = 6.02 \times 10^{23} \text{ mol}^{-1}$)

- A)** 1.20×10^{24} **B)** 3.01×10^{22} **C)** 3.01×10^{23} **D)** 6.02×10^{23} **E)** 9.03×10^{23}

41. A sealed elastic balloon at constant pressure holds air at 27 °C. It is warmed to 327 °C. Its volume:

- A)** increases about 1.1-fold
- B)** stays the same
- C)** increases about 12-fold
- D)** halves
- E)** doubles

42. Which atom has the largest radius?

- A)** Mg **B)** K **C)** Na **D)** Cl **E)** Al

43. A white crystalline solid has a high melting point, shatters when struck, and conducts electricity only when molten or dissolved. It is most likely:

- A)** a network covalent solid such as quartz
- B)** a covalent molecular solid such as wax
- C)** a metal such as copper
- D)** graphite
- E)** an ionic compound such as NaCl

44. Ethanoic acid reacts with ethanol in the presence of a few drops of concentrated H_2SO_4 . The organic product is:
- A) ethyl ethanoate
 - B) ethanal
 - C) ethanoic anhydride
 - D) sodium ethanoate
 - E) ethanol
45. Which single observation distinguishes ethene from ethane?
- A) Both burn in air with a luminous flame.
 - B) Ethane is a gas; ethene is a liquid at room temperature.
 - C) Ethene dissolves freely in water.
 - D) Ethane contains a carbon-carbon double bond.
 - E) Ethene decolourises bromine water; ethane does not (under these conditions).
46. For the exothermic equilibrium $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$, raising the temperature at constant pressure:
- A) increases both the rate and the equilibrium yield
 - B) decreases the rate but increases the yield
 - C) affects neither rate nor yield
 - D) increases the rate but decreases the equilibrium yield
 - E) increases the rate and leaves the yield unchanged
47. A solution contains 0.10 mol of glucose dissolved in 500 g of water. Its molality is:
- A) 0.05 mol/kg
 - B) 0.10 mol/kg
 - C) 0.20 mol/kg
 - D) 0.50 mol/kg
 - E) 2.0 mol/kg

Section D - Mathematics (Q48-54)

48. Solve the inequality $5 - 3x \geq 14$.
- A) $x < -3$
 - B) $x \leq -3$
 - C) $x \leq 3$
 - D) $x \geq 3$
 - E) $x \geq -3$
49. What is the domain of $f(x) = \sqrt{(4 - x) / (x - 2)}$?
- A) $x \leq 4$ with $x \neq 2$
 - B) $x \leq 4$ with $x \neq 4$
 - C) $x \leq 4$
 - D) $x < 4$
 - E) all real numbers
50. What is the value of $\log_3 81$?
- A) 3
 - B) 4
 - C) 9
 - D) 27
 - E) 81
51. Three fair coins are tossed. What is the probability of getting at least one head?
- A) $1/2$
 - B) $1/8$
 - C) $2/3$
 - D) $3/8$
 - E) $7/8$
52. A straight ramp 10 m long makes a 60° angle with the horizontal ground. How far along the ground does the ramp span (horizontal distance between its two ends)? ($\cos 60^\circ = 0.5$, $\sin 60^\circ = 0.87$)
- A) 2.5 m
 - B) 5 m
 - C) 6.7 m
 - D) 8.7 m
 - E) 10 m
53. If $f(x) = 2x - 3$ and $g(x) = x^2$, what is $f(g(2))$?
- A) 1
 - B) 2
 - C) 4
 - D) 5
 - E) 8
54. Simplify $(2^5 \times 2^3) \div 2^4$.
- A) 4
 - B) 8
 - C) 16
 - D) 32
 - E) 64

Section E - Physics (Q55-60)

55. A 6 V battery drives a current through a $2\ \Omega$ resistor connected in series with a parallel pair made of a $6\ \Omega$ and a $3\ \Omega$ resistor. What current flows from the battery?
- A) 0.5 A
 - B) 1.0 A
 - C) 1.5 A
 - D) 3.0 A
 - E) 6.0 A

- 56.** A kettle draws 4.6 A from a 230 V supply. Its power consumption is closest to:
A) 50 W **B)** 230 W **C)** 1060 W **D)** 2300 W **E)** 4600 W
- 57.** Equal masses of aluminium (specific heat 900 J/(kg·K)) and copper (specific heat 385 J/(kg·K)) receive the same quantity of heat. Which statement is correct?
A) Copper's temperature rises about 2.3 times more than aluminium's.
B) Both rise by the same amount.
C) Copper's temperature does not change.
D) It cannot be determined without knowing the exact heat supplied.
E) Aluminium's temperature rises more than copper's.
- 58.** A car travelling at 20 m/s brakes uniformly and comes to rest in 4.0 s. The magnitude of its deceleration is:
A) 0.2 m/s² **B)** 4 m/s² **C)** 5 m/s² **D)** 16 m/s² **E)** 20 m/s²
- 59.** A velocity-time graph shows: a linear rise from 0 to 4 m/s during the first 4 s, a constant 4 m/s for the next 4 s, then a linear fall back to 0 over the final 4 s. What total distance is covered?
A) 16 m **B)** 24 m **C)** 32 m **D)** 40 m **E)** 48 m
- 60.** A block floats stationary in water. Which statement must be true?
A) The buoyant force is zero.
B) The density of the block is greater than that of water.
C) The mass of displaced water equals the volume of the block.
D) The weight of the displaced water equals the weight of the block.
E) The block displaces a volume of water equal to its own volume.

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	D	2	E	3	A	4	C	5	C
6	B	7	A	8	B	9	B	10	B
11	D	12	E	13	E	14	D	15	C
16	A	17	D	18	C	19	A	20	E
21	D	22	E	23	A	24	A	25	C
26	B	27	C	28	D	29	C	30	B
31	E	32	B	33	C	34	A	35	D
36	C	37	D	38	C	39	C	40	C
41	E	42	B	43	E	44	A	45	E
46	D	47	C	48	B	49	A	50	B
51	E	52	B	53	D	54	C	55	C
56	C	57	A	58	C	59	C	60	D